



- Constituent College of JSS Science and Technology University
- Approved by A.I.C.T.E
- Governed by the Grant-in-Aid Rules of Government of Karnataka
- Identified as lead Institution for World Bank Assistance under TEQIP Scheme



DEPARTMENT OF HUMANITIES
SCHEME & SYLLABUS
First Semester
(From the Academic Year 2024-25)

DEPARTMENT	Humanities						
Course Code	24HU120 /24HU220	Total Credits	1	Course Type	Engineering Science Course – ESC		
Course Title	INNOVATION & DESIGN THINKING						
Teaching Learning Process		Contact Hours/week	Credits	Assessment in Weightage and marks			
	Lecture	2	1		CIE	SEE	Total
	Tutorial	0	-	Weightage %	40	60	----
	Practical	0	-	Maximum Marks	20	30	50
	Total	2	1	Minimum Marks *	10	13	23

***Note:** For passing, the student has to score a minimum of 23 marks (CIE + SEE = 10+13 or 11 +12).

Course Overview: This course provides an introduction to the need of Innovations and design thinking with necessary basic concepts, types, tools and techniques involved in practice to cater to the engineering needs.

Course Objective:

- To explain the concept of innovations and its types, needs, implementation and protection.
- To apprise the importance of design thinking for strategic innovations.

Course Outcome:

CO#	Course Outcome	Highest Level of Cognitive Domain
CO1	Comprehensively classify, compare, discuss and illustrate the types of innovations with examples.	L2
CO2	Explain the models of innovations with examples and illustrate applications related to past, present and quantify future needs.	L2
CO3	Describe methods of protection to innovations and develop conceptual design of products/services relevant to respective branch of engineering.	L3
CO4	Illustrate the need, types, tools and implementation of design thinking and elucidate the collaborated design thinking.	L2
CO5	Explain the impact of design thinking for strategic innovations.	L2

L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze, L5 – Evaluate, L6 - Create

Course Articulation:

POs → COs ↓	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	-	-	3	-	-	-	-	-	-	-	-	1
CO2	-	-	3	-	-	-	-	-	-	-	-	1
CO3	-	-	3	-	-	-	-	-	-	-	-	1
CO4	-	-	3	-	-	-	-	-	-	-	-	1
CO5	-	-	3	-	-	-	-	-	-	-	-	1

High – 3, Medium – 2, Low – 1

Course Content:

UNIT No.	Content	Hours
UNIT 1	INTRODUCTION TO INNOVATION: Innovation – Introduction, Definition, Inventions, discovery and Innovations, Historical developments of inventions leading to innovations, Classification Of Innovation – Based On Category - Product Innovation, Process Innovation, Service Innovation, Marketing Innovation, Organizational Innovation, Based On Overall Global Needs – Technological Innovation, Social Innovation, Based On the Degree Of Novelty - Incremental Innovations, Radical Innovations, Systemic /Systematic Innovations, Breakthrough Innovation. Creativity and innovation	7

UNIT 2	INNOVATION NEEDS AND MODELLING: Need for innovation - Importance of innovation- business Needs, Societal Needs, Technological, Scientific and other needs, Innovations of the past, present and the future. Process of Brainstorming. Advantages and disadvantages of innovation. Types and Importance of models, TEDOC and SCAMPER - models of Innovation.	7
UNIT 3	PROTECTION OF INNOVATION: Safeguarding innovation, methods of protection and IPR - Patent, Trademark, Industrial design, Geographical Indication, Copyright. WIPO	4
UNIT 4	DESIGN THINKING & ITS TOOLS: Introduction to design thinking, Theory and practice in Design thinking, Implementation, MVP & Prototyping, Scenario based Prototyping. Tools for Design thinking, Collaboration in distributed Design.	4
UNIT 5	DESIGN THINKING FOR STRATEGIC INNOVATIONS: Growth – Story telling representation – Strategic Foresight - Change – Sense Making - Maintenance Relevance – Value redefinition - Extreme Competition – experience design - Standardization – Humanization - Creative Culture – Rapid prototyping, Strategy and Organization – Business Model design.	4

Text Books:

1. “The Houdini Solution. Put Creativity and Innovation to Work by Thinking INSIDE THE BOX”, By Ernie Schenck, McGraw Hill, New York 2007.
2. Idris Mootee, "Design Thinking for Strategic Innovation: What They Can't Teach You at Business or Design School", John Wiley & Sons 2013.
3. The Little Black Book of Innovation: How It Works, How to Do It? - Scott D. Anthony, Harvard Business School Press – 2011.
4. John.R.Karsnitz, Stephen O'Brien and John P. Hutchinson, “Engineering Design”, Cengage learning (International edition), Second Edition, 2013.

Reference Books:

1. Reverse Innovation: Create Far From Home, Win Everywhere - Vijay Govindarajan, Chris Trimble and Indra K. Nooyi - Harvard Business Press Books – 2012.
2. Innovation. - Growth Engine for Nation -. Nice Buzzword but Often Misunderstood. Compiled By. Dr. Rajiv V. Dharaskar. Ph.D.
3. Hasso Plattner, Christoph Meinel and Larry Leifer (eds), "Design Thinking: Understand – Improve – Apply", Springer, 2011.

Web/Digital resources:

1. www.mindtools.com
2. www.quicksprout.com
3. www.mindtools.com

Evaluation Scheme:**Continuous Internal Evaluation – CIE**

Event	Event Type	Marks Allotted
CIE – 1	Written Test – 1	15
CIE – 2	Event	10
CIE – 3	Written Test – 2	15

Note:

1. The written test 1 & 2 (CIE – 1 & 3) both will be conducted for 15 marks each in 1 hour duration.
2. The Event (CIE – 2) will be conducted for 10 marks.
3. The Event (CIE – 2) will be skill based assessment such as Seminars / Technical talks / Case study / hands-on activity / Mini projects / Sci-tech activity / Data analysis.
4. A student must score on an average of 50% i.e., 20 marks out of 40 from all the events (CIE - 1, 2, 3) to gain the eligibility to appear for SEE.
5. The final CIE (40 Marks) will be reduced proportionately to 20 marks.
6. The eligibility mark is 10 to appear for SEE.

Semester End Examination – SEE

Event	Event Type	Marks Allotted	Duration
SEE	Written Examination	60	2 Hours

Note:

1. The SEE will be conducted for 60 marks and the marks. A student must score a minimum of 24 marks out of 60 in the SEE to pass.
2. SEE Question paper will be set for 60 marks and will have two parts (Part-A & Part-B) with equal weightage / marks distribution of 30 marks for each PART A and PART B. Questions under Part-A are compulsory and questions under Part-B will have internal choices.
3. The final SEE will be reduced to 30 marks.
4. The passing marks in SEE is 12 marks out of 30 marks.
5. To pass the course, the student has to score a minimum of 23 marks (CIE + SEE = 10+13 or 11 +12).